Bryan Pham

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. Degree: Mechanical Engineering (Transfer GPA: 4.0) — *June. 2026 – May 2028*

Experience

Design & Manufacturing Engineering Intern — June. 2026 – Present
Mission College | https://abgsccc-website.vercel.app/ — *Santa Clara, CA*

- Designed and optimized the airframe for a high-powered model rocket in Onshape and SolidWorks, including a tilt and roll control system, ensuring structural integrity, steady component implementation, and at least 1.0 cal of stability in the vehicle
- Conducted 3000+ hours of computational fluid dynamics (CFD) simulations in SimScale to validate aerodynamic performance, drag, pressure distribution, and stability prior to fabrication
- Engineered an active-control tilt/roll system using micro-servos, reading acceleration and angular velocity data from an IMU module, and translating changes to canard deflections with an ESP32 and Raspberry Pi 5 system
- Prototyped and implemented live video capture with a drone runcam to analyze interactive differences between CFD simulation and live runs while reducing drag

Associated Student Government Senator — Aug. 2025 – May 2026
Mission College — *Santa Clara, CA*

- Represented student interests in discussions with college administrators and campus leadership
- Advocated for policies & programs that improve student engagement, academic success, and resource accessibility
- Participated in institutional leadership retreat addressing campus-wide academic and operational initiatives

Projects

Aside AI | *1st DeepGram Track, UC Berkeley AI Hackathon | DeepGram, Redis, QNX, Python, Microcomputers*

- Built a clip-on camera and mic that narrates the user's surroundings in real time through different AI personalities
- Developed a modular system using a Raspberry Pi (QNX 8.0) capture device, Python coordination, and React Native app, achieving live end-to-end narration in 1–2 seconds

Mission Launch Rocketry | *OnShape, 3D Printing, Microcomputers, Microcontrollers*

- Founded and served as President of college rocketry club, overseeing 52 members, managing budget, coordinating project/promotional events, and directing design-build-launch projects from concept through flight
- Integrated dual-deployment recovery system in a two-stage high power rocket (drogue and main parachutes) with staged separation sequencing from EasyMini and EasyMega flight computers
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Recco | *Y Combinator AI Growth Hackathon | Computer Vision, Deepgram, Fiber, Convex, CV Service*

- Built a camera-first iOS networking assistant that identifies people at events in real time by fusing on-device face tracking with cloud vision, identity lookup, and voice control
- Engineered a SwiftUI + AVFoundation camera pipeline with Apple Vision face tracking, a target-lock reticle, and an AR overlay that resolves the person nearest screen center on voice or text command

RollAway | *1st Beginner Track, MLH x Digital Ocean AI for Social Good Hackathon | React, TypeScript, UIX, FastAPI*

- A map-first location-intelligence and permit-planning app for mobile food vendors in San Francisco that ranks legal, low-competition places to set up for a chosen time window and simplifies the permit acquisition process
- Built the permit checklist functionality using React and TypeScript on a DigitalOcean serverless backend and implemented a Zustand-driven UI that consolidates user-ingested data into auto-filled forms with manual text entry to improve accessibility.

Technical Skills

Languages : C, C++, Python, TypeScript, JavaScript, MATLAB, Swift, HTML/CSS
CAD: SolidWorks, Onshape, AutoCAD, Fusion360, MecAgent, Zoo.dev
Manufacturing: 3D Printing, Soldering, Microcontrollers, SimScale, Microcomputers, Computer Vision, PSpice, LTSpice, Oscilloscope, Function Generator
Tools & Platforms: Claude Code, Codex, Cursor, Git/Github, Github Actions, Redis, Mapbox